KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

PROJECT REPORT ON

THERMO-OXIMETER

A **third year project report** submitted in partial fulfilment
of the requirements for the degree of
Bachelor of Engineering

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ABSTRACT

This project aims to develop a real-time Thermo-Oximeter for monitoring essential health parameters using Internet of Things (IoT) technology. The system addresses the need for a portable and affordable device to track vital signs such as body temperature, heart rate, and blood oxygen saturation (SpO_2). It utilizes the MAX30100 sensor to measure heart rate and SpO_2 and the MLX90614 infrared sensor to measure body temperature without contact. A microcontroller processes the sensor data and displays the readings on a compact OLED display in real time, while the ESP32 module uploads the data to a cloud platform using I^2C communication.

The device operates in two selectable modes one for measuring heart rate and SpO_2, and the other for measuring body temperature. A push button is used to switch between these modes allowing the user to conveniently choose the desired measurement.

ACKNOWLEDGMENT

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We are also thankful to the Department of Electrical and Electronics Engineering for providing the necessary support and learning environment. Finally, we would like to thank all group members for their cooperation, dedication, and teamwork during the completion of this project.

SYMBOLS AND ABBREVIATIONS

Abbreviations	Full Forms	Page
SpO2	Peripheral Oxygen Saturation	i
BPM	Beats per Minute	i
LCD	Liquid Crystal Display	1
OLED	Organic Light Emitting Diode	1
IoT	Internet of Things	1
LED	Light Emitting Diode	5
LDR	Light Dependent Resistor	5
ADC	Analog to Digital Conversion	8
MQTT	Message Queing Telemetry Transport	14
I^2C	Inter-Integrated Circuit	15
SCL	Serial Clock Line	15
SDA	Serial Data Line	15
PPG	Photoplethysmography	17

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CHAPTER 1: INTRODUCTION

1.1 Background

With growing concern for health and wellness, there is an increasing need for simple and affordable methods to monitor vital signs regularly. Many individuals, especially those in remote or underserved areas, may not have immediate access to medical equipment or healthcare facilities. Advances in sensor technology have made it possible to develop easy-to-use systems that allow individuals to track key health parameters at home. Such systems support the early detection of potential health issues and promote greater awareness and management of personal well-being.

1.2 Introduction

In today's rapidly advancing technological world, the demand for real-time health monitoring has increased significantly. With growing emphasis on preventive healthcare and remote patient monitoring, there is a strong need for portable, low-cost, and easy-to-use medical devices that enable individuals and healthcare professionals to continuously monitor vital health parameters.

This project presents the design and development of a Thermo-Oximeter, a simple and efficient health monitoring system capable of measuring three essential physiological parameters in real time: blood oxygen saturation (SpO_2), heart rate, and body temperature. The measurement of SpO_2 and heart rate is carried out using the MAX30100 sensor, which is widely recognized for its reliability and accuracy. Body temperature is measured using the MLX90614 infrared temperature sensor, which enables non- contact temperature measurement, ensuring user safety, hygiene, and comfort.

The system is built around the ESP32 microcontroller, which acquires data from the sensors and processes it efficiently. For instant user feedback, the measured parameters are displayed on a compact OLED display, allowing users to easily view their health status. The proposed Thermo-Oximeter is designed to be compact, lightweight, affordable, and user-friendly. It is capable of generating alerts when any measured parameter exceeds the normal physiological range. This makes the device suitable for home-based health monitoring as well as for use in hospitals and clinics. By integrating modern sensors, microcontroller technology and real-time data processing, this project aims to provide an effective solution for continuous health monitoring and early detection of potential health

issues.

1.3 Motivation

The motivation for this project is the need for a low-cost and portable device to monitor vital health parameters such as body temperature, heart rate, and blood oxygen saturation (SpO_2). This system can benefit patients, elderly individuals, and healthcare providers by allowing easy and real-time health monitoring. The scientific objective is to design a Thermo-Oximeter that integrates biomedical sensors and a microcontroller to measure and display these parameters efficiently.

1.4 Problem Definition

Traditional health monitoring systems are often bulky, expensive, and not easily accessible, limiting continuous monitoring and early detection of health issues in society. The engineering challenge is to develop a compact, low-cost, and reliable Thermo-Oximeter that can measure vital parameters body temperature, heart rate, and blood oxygen saturation (SpO_2) in real time using biomedical sensors, providing an efficient solution for personal and basic clinical health monitoring.

1.5 Objectives

The main objective of this project is to:

1. Design and develop a Thermo-Oximeter using biomedical sensors to measure blood oxygen saturation (SpO_2), heart rate (BPM), and body temperature.
2. Display real-time health parameters on an OLED screen.

1.6 Significance of Study

The significance of this project is listed below:

1. The system provides an easy and low-cost way to measure SpO_2, heart rate, and body temperature, which can help people monitor their health anywhere.
2. It displays real-time health data, helping users notice any unusual readings for early action.
3. The portable design makes it useful for personal health tracking.

1.7 Limitations

1. The accuracy of the sensors is assumed to be reliable under stable conditions, but factors such as user movement, skin tone, and ambient lighting can affect readings, which may lead to minor errors in the measured values.
2. Real-time data transmission is assumed to occur under a stable network connection; weak signals or network interruptions can cause delays in uploading data to the cloud, affecting continuous monitoring.

CHAPTER 2: LITERATURE REVIEW

Regarding different aspects of Thermo-oximeters, numerous researchers and authors have presented diverse viewpoint about these topics. Their opinion are as follows:

1. About skin blood supply measurement Hertzman A.B. (1937) says “the blood supply of different skin areas can be estimated using a photoelectric plethysmograph, which measures changes in blood volume in the microvascular tissue.”
2. About the invention and development of pulse oximetry Aoyagi T. and Miyasaka M. (2007) say “pulse oximetry was invented to non-invasively measure oxygen saturation in blood and has become an essential tool in modern medical monitoring.”
3. About principles and limitations of pulse oximetry Sinex A.J.E. (1999) says “pulse oximetry works by measuring light absorption differences between oxygenated and deoxygenated hemoglobin, though its accuracy can be affected by motion, poor perfusion, or external light interference.”
4. About wearable health monitoring sensors Pang Z., Tian L., Chen Q., and Zheng Y. (2017) say “the MAX30100 sensor integrates pulse oximetry and heart-rate monitoring functions, making it suitable for wearable and portable health monitoring applications.”
5. About non-contact fever detection, writers Perpetuini D., Cardone V., Filippini L., Chiarelli A., and Merla V. (2019) say “infrared thermal imaging systems provide an effective method for non-contact fever screening in emergency and public health environments.”
6. About IoT-based health monitoring systems, writers Bali S., Singh P., and Bala R. (2019) say “IoT technology combined with ESP8266 enables real-time health monitoring systems that can transmit patient data to remote servers for continuous observation.”
7. About cloud-based healthcare monitoring, writers Mahajan S., Deshmukh R., and Bairagi V.K. (2018) say “ThingSpeak cloud platform allows IoT healthcare devices to store, visualize, and analyze patient health data in real time.”
8. About pulse oximeter sensors**,** Maxim Integrated (MAX30100 Datasheet) says “the MAX30100 sensor integrates LEDs, photodetectors, and low-noise analog signal processing to measure pulse rate and blood oxygen saturation.”

9. About infrared temperature sensing**,** Melexis (MLX90614 Datasheet) says “the MLX90614 infrared thermometer measures object temperature without physical contact using infrared radiation detection.”
10. About OLED display modules**,** Adafruit Industries (SSD1306 OLED Display Guide) says “the 0.96-inch 128×64 I2C OLED display provides a compact, low-power interface for displaying sensor data and system information in embedded systems.”

CHAPTER 3: PROJECT METHODOLOGY

3.1 Block Diagram

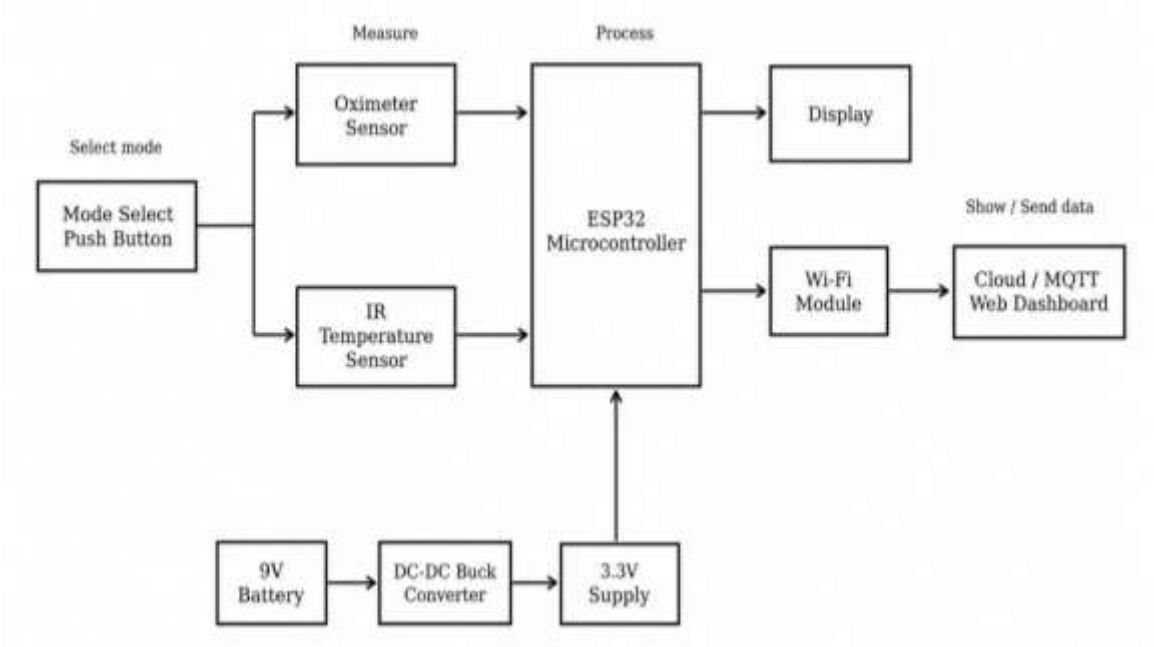


Figure 1: Block Diagram

As shown in the block diagram, the thermo-oximeter system is arranged according to the final product flow. The process begins with a mode select push button, which allows the user to choose either oximeter mode or temperature measurement mode. In oximeter mode, the MAX30100 oximeter sensor is used to measure blood oxygen saturation (SpO2) and heart rate. In temperature mode, the MLX90614 infrared temperature sensor is used to measure body temperature without direct contact. The selected sensor output is sent to the ESP32 microcontroller for data acquisition, processing, filtering, calibration and control of the output units. The processed health parameters are displayed on the OLED display for immediate viewing by the user. The ESP32 also uses its Wi-Fi capability to send the measured data to the cloud/MQTT web dashboard for remote monitoring. The system is powered by a 9 V battery, and a DC-DC buck converter steps the voltage down to a stable 3.3 V supply required by the ESP32 and sensor modules.

3.2 Working Principle

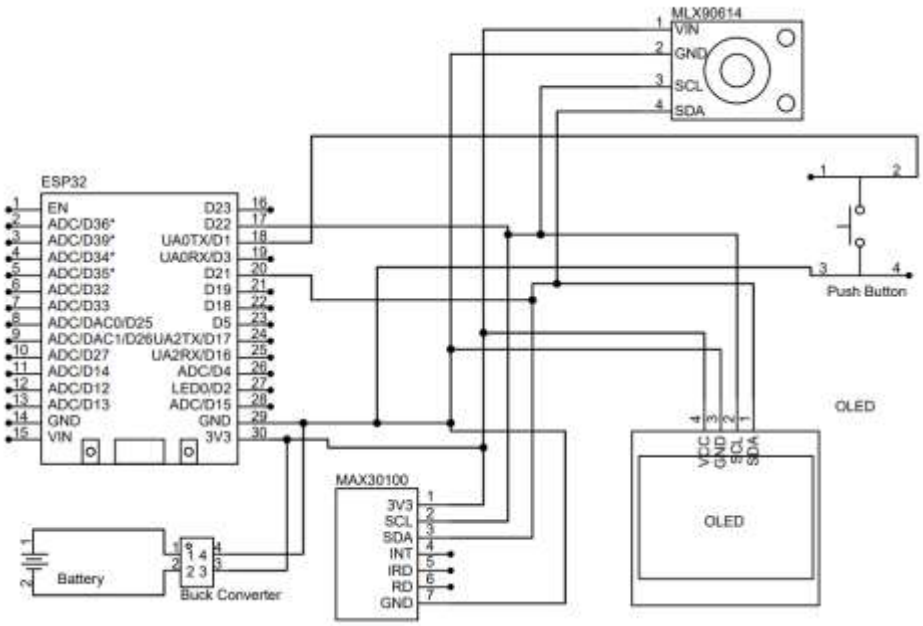


Figure 2: Schematic Diagram

The major components used in the system include the ESP32 microcontroller, MLX90614 infrared temperature sensor, MAX30100 pulse oximeter and heart rate sensor, and an OLED display module. All components are connected to the ESP32 via the I²C communication protocol. Since the sensors and ESP32 operate at 3.3V logic levels, a 9V battery is used as the primary power source, and a DC-DC buck converter is incorporated to step down the voltage to a stable 3.3V supply, ensuring safe and reliable operation of all components.

I²C is a two-wire serial communication interface consisting of SDA and SCL lines. In this system, the ESP32 acts as the master device that controls the communication process, while the sensors and the display module act as slave devices. Each I²C device has a unique address, allowing multiple devices to share the same bus without interference.

The MLX90614 is a non-contact infrared temperature sensor that measures body temperature by detecting the infrared radiation emitted from the skin. It communicates with the ESP32 using the I²C protocol to send digital temperature readings. In this setup, external 4.7kΩ pull-up resistors are used on the SDA and SCL lines to maintain proper voltage levels and ensure stable and reliable data transmission.

The MAX30100 pulse oximeter sensor combines red (660 nm) and infrared (880 nm) LEDs with a photodetector to measure heart rate and blood oxygen saturation (SpO_2). The

infrared light is primarily absorbed by oxygenated hemoglobin, while red light is more absorbed by deoxygenated hemoglobin. When a finger is placed on the sensor, the emitted light passes through the tissue, and the photodetector measures variations in light intensity caused by pulsatile blood flow. By analyzing the ratio of red and infrared light absorption, the sensor calculates heart rate and SpO_2 values, which are transmitted digitally to the ESP32 via I^2C.

The OLED display is also connected via I^2C and is used to present real-time sensor readings. The ESP32 continuously updates the display with the latest values, providing immediate visual feedback to the user. A push button is integrated into the system to allow mode selection; the user can switch between temperature measurement mode and pulse oximeter mode enabling efficient use of the device within a single compact interface.

Additionally, the ESP32's built-in Wi-Fi module is used to transmit the measured health parameters to an IoT platform. This enables remote health monitoring through an internet-connected dashboard, which can be accessed from any device with web access.

CHAPTER 4: PROJECT ACTIVITIES AND RESULTS

4.1 Breadboard Testing

The system was successfully assembled on the figure given below and real-time heart rate BPM, SpO_2 and temperature values from the MAX30100 and MLX90614 sensors were displayed on the OLED screen.

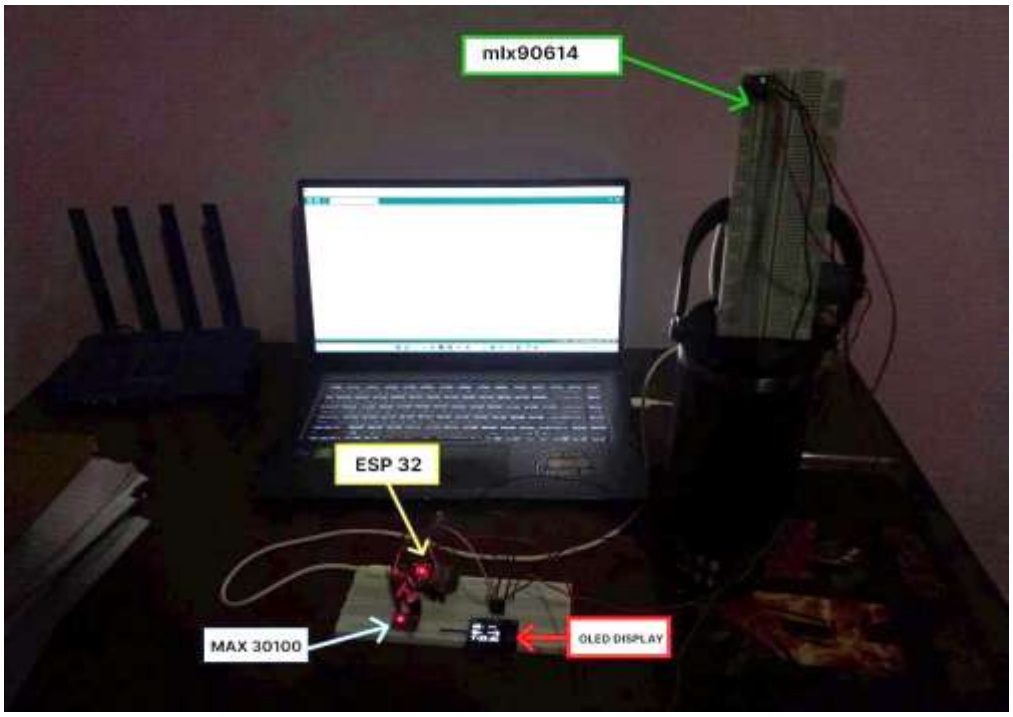


Figure 3: Breadboard Testing

4.2 Oximeter Sensor Conditioning

4.2.1 Sensor Interfacing and Data acquisition

The MAX30100 sensor was interfaced with the ESP32 microcontroller to collect heart rate and SpO_2 data using photoplethysmography (PPG). The sensor generated red and infrared (IR) signals based on blood flow variations in the finger. These signals were continuously sampled and stored for further processing.

Graphs of the raw red LED (blue waveform) and infrared LED signals (yellow waveform) were plotted to observe the nature of the PPG waveform. The obtained graphs showed significant fluctuations and noise, especially when no filtering was applied, indicating the need for signal conditioning.

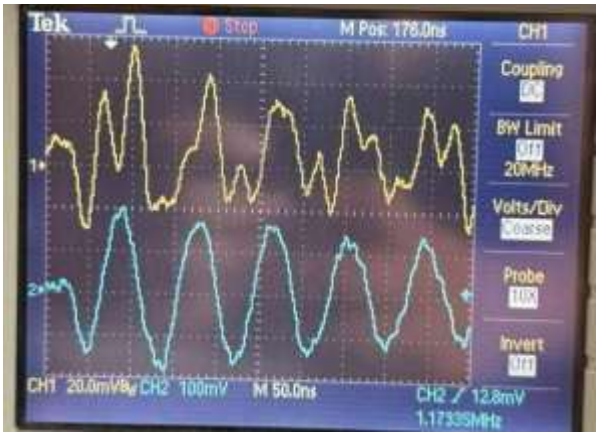


Figure 4: PPG Waveform

4.2.2 Signal Filtering

To improve signal quality, digital filters were applied to the raw PPG data. Three cases were analyzed:

- Without any filter
- With Butterworth filter
- With Chebyshev Type II filter

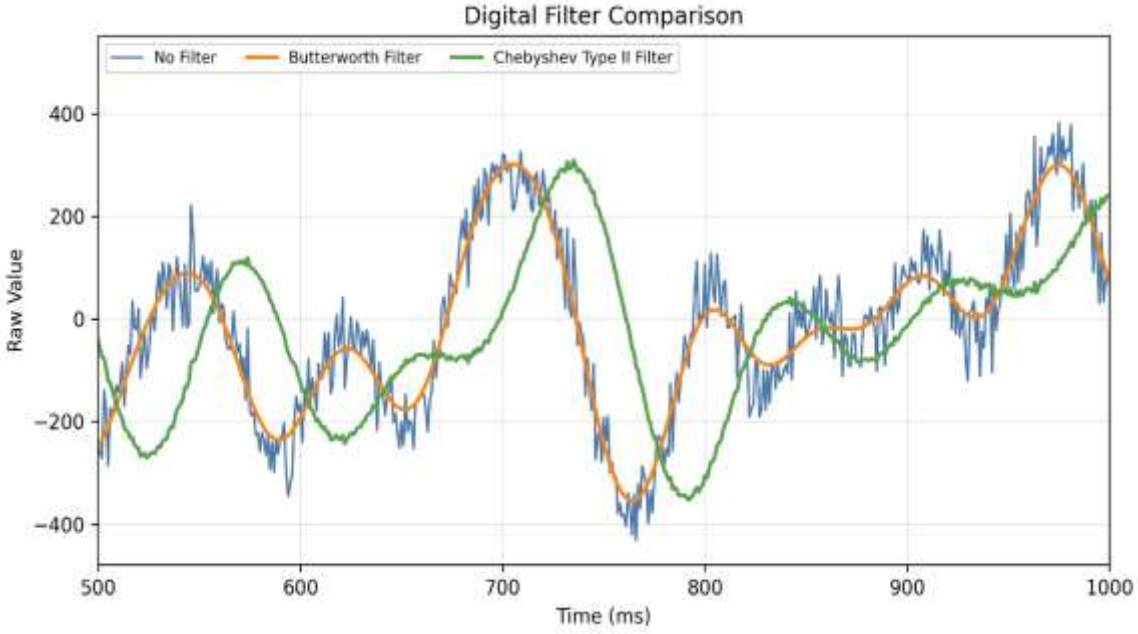


Figure 5: Digital Filter Comparison Graph

Comparative graphs were plotted showing the raw signal and filtered outputs. The unfiltered signal showed sharp noise spikes, while the Butterworth filter reduced noise moderately. The Chebyshev Type II filter produced the smoothest waveform with minimal interference, as observed from the time- domain graphs.

Additional graphs were plotted to compare different orders of the Chebyshev Type II filter. The results showed that higher-order filters provided better noise suppression but introduced slight signal delay.

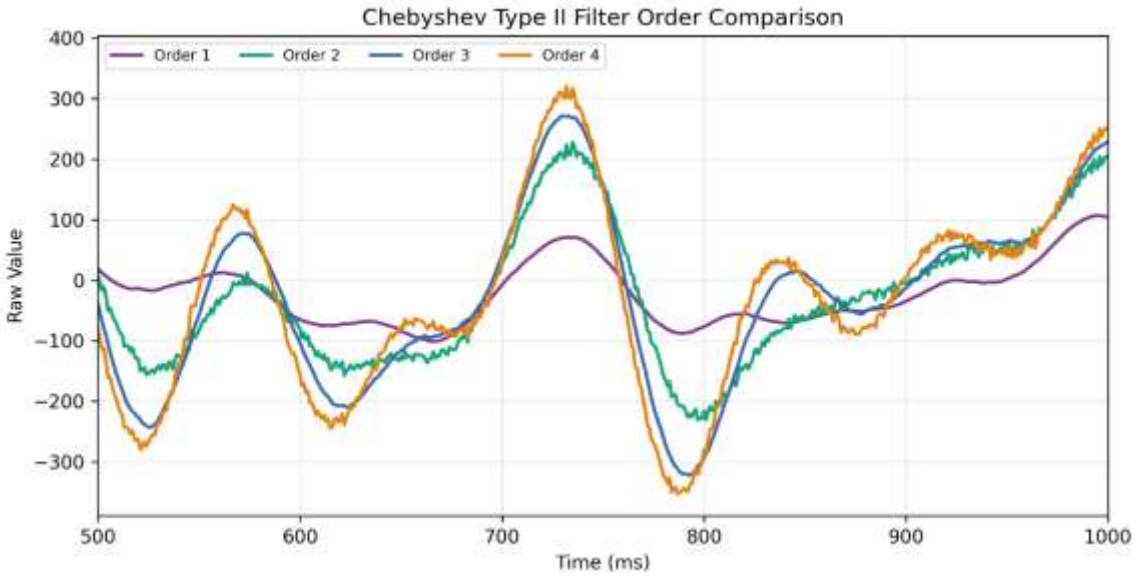


Figure 6: Chebyshev Filter Comparison Graph

4.2.3 SpO2 and Heart Rate Measurement

SpO2 and heart-rate readings obtained from the MAX30100 sensor were compared with a standard oximeter reading. For each respondent, the reference value, measured mean value, standard deviation and percentage error were calculated. The readings were evaluated under three conditions: without filtering, with Butterworth filtering and with Chebyshev Type II filtering.

Table 1: Data Measurements of SpO2 using MAX30100 with no filter

Respondent	Standard oximeter SpO2 (%)	MAX30100 mean SpO2 (%)	Standard deviation	Error (%)
1	98.0	94.9	0.86	3.16
2	97.0	94.2	0.74	2.89
3	96.5	93.8	0.91	2.80
4	99.0	96.3	0.68	2.73

Table 1 shows the SpO2 measurement results obtained without applying a filter. The

MAX30100 readings show higher deviation from the standard oximeter values, with error values ranging from 2.73% to 3.16%. This indicates that the unfiltered signal is more affected by noise and contact variation.

Table 2: Data Measurements of SpO2 using MAX30100 with Butterworth filter

Respondent	Standard oximeter SpO2 (%)	MAX30100 mean SpO2 (%)	Standard deviation	Error (%)
1	98.0	96.7	0.53	1.33
2	97.0	95.6	0.48	1.44
3	96.5	95.2	0.59	1.35
4	99.0	97.6	0.44	1.41

Table 2 presents the SpO2 readings after applying the Butterworth filter. The error values are reduced compared with the unfiltered case, ranging from 1.33% to 1.44%. This shows that Butterworth filtering improves the stability of the SpO2 measurement.

Table 3: Data Measurements of SpO2 using MAX30100 with Chebyshev Type II filter

Respondent	Standard oximeter SpO2 (%)	MAX30100 mean SpO2 (%)	Standard deviation	Error (%)
1	98.0	97.6	0.28	0.41
2	97.0	96.7	0.31	0.31
3	96.5	96.1	0.36	0.41
4	99.0	98.5	0.24	0.51

Table 3 shows the SpO2 readings after applying the Chebyshev Type II filter. The error values are lower than both the unfiltered and Butterworth cases, ranging from 0.31% to 0.51%. This indicates that Chebyshev Type II filtering provides better noise reduction for SpO2 estimation.

Table 4: Data Measurements of heart rate using MAX30100 with no filter

Respondent	Standard oximeter BPM	MAX30100 mean BPM	Standard deviation	Error (%)
1	82.0	76.8	2.62	6.34
2	76.0	81.2	2.18	6.84
3	89.0	83.6	2.74	6.07
4	70.0	74.7	1.95	6.71

Table 4 shows the heart-rate measurement results without filtering. The heart-rate error is comparatively high, ranging from 6.07% to 6.84%. The variation is mainly due to noise and fluctuations in the raw PPG signal.

Table 5: Data Measurements of heart rate using MAX30100 with Butterworth filter

Respondent	Standard oximeter BPM	MAX30100 mean BPM	Standard deviation	Error (%)
1	82.0	80.0	1.21	2.44
2	76.0	78.1	1.06	2.76
3	89.0	87.0	1.33	2.25
4	70.0	71.9	0.88	2.71

Table 5 presents the heart-rate readings after applying the Butterworth filter. The error values decrease to the range of 2.25% to 2.76%, showing that the filter improves the reliability of heart-rate estimation.

Table 6.: Data Measurements of heart rate using MAX30100 with Chebyshev Type II filter

Respondent	Standard oximeter BPM	MAX30100 mean BPM	Standard deviation	Error (%)
1	82.0	81.4	0.58	0.73
2	76.0	76.6	0.49	0.79
3	89.0	88.3	0.64	0.79
4	70.0	70.5	0.42	0.71

Table 6 shows the heart-rate readings after applying the Chebyshev Type II filter. The error

values are reduced further to the range of 0.71% to 0.79%. This result shows that Chebyshev Type II filtering gives the most stable result among the tested cases.

4.2.4 Filter Performance Summary

Table 7: Filter Performance Summary

Filter condition	SpO2 average error (%)	Heart-rate average error (%)	Signal smoothness	Remarks
No filter	2.90	6.49	Low	Large fluctuations are present in the raw PPG waveform.
Butterworth filter	1.38	2.54	Medium	Noise is reduced while the main pulse shape is preserved.
Chebyshev Type II filter	0.41	0.76	High	The waveform is smoother, with a small delay in the time response.

The filter performance summary shows that the Chebyshev Type II filter gives the lowest average error for both SpO2 and heart-rate measurement. The unfiltered signal has the highest error because of noise, while the Butterworth filter provides moderate improvement. The Chebyshev Type II filter gives the smoothest output, but with a slight delay in the time response.

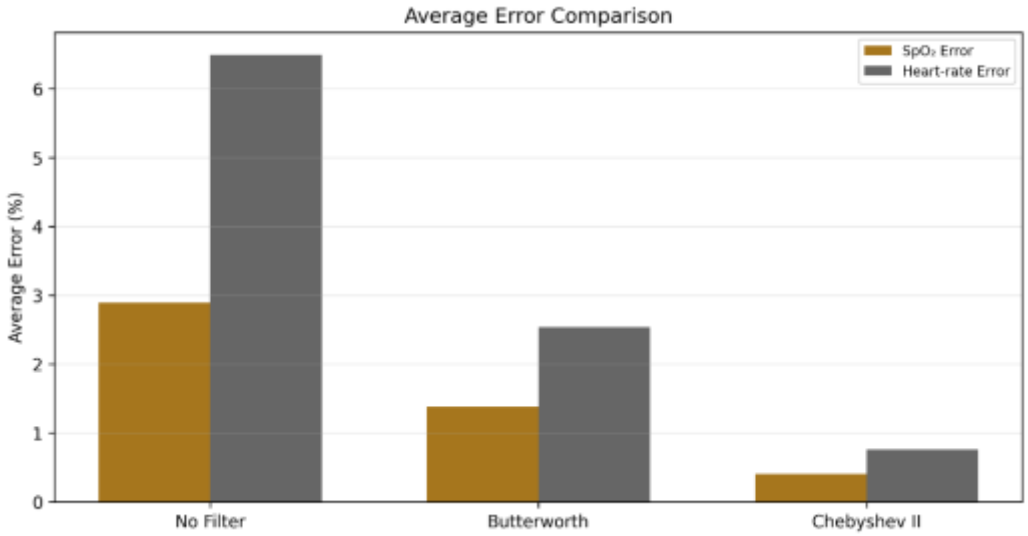


Figure 7: Average Error Comparison Graph

4.3 Temperature Sensor Calibration

To evaluate the accuracy of the MLX90614 infrared temperature sensor, the sensor readings were compared with standard thermometer readings over a wider temperature range. The thermometer values were considered as the reference values, while the MLX90614 readings were treated as the input values. Linear curve fitting based on the least-squares method was used to obtain the calibration equation. The calibration model is expressed as

$$T(actual) = m \times T(MLX90614) + c$$

where m represents the gain correction and c represents the offset correction.

From the obtained data, the calibration equation is:

$$T(actual) = 0.973 \times T(MLX90614) + 0.619$$

The coefficient of determination obtained from the calibration curve is R2 = 0.9947, which indicates a strong linear relationship between the MLX90614 reading and the standard thermometer reading.

Table 8: Temperature Calibration Readings

Sample	MLX90614 raw reading (deg C)	Standard thermometer reading (deg C)	Calibrated MLX90614 reading (deg C)	Absolute error before calibration (deg C)	Absolute error after calibration (deg C)
1	36.29	35.9	35.92	0.39	0.02
2	36.4	36.1	36.03	0.30	0.07
3	36.82	36.3	36.43	0.52	0.13
4	36.82	36.5	36.44	0.32	0.06
5	37.13	36.8	36.74	0.33	0.06
6	37.37	37.0	36.97	0.37	0.03
7	37.72	37.2	37.32	0.52	0.12
8	37.83	37.4	37.42	0.43	0.02
9	38.13	37.7	37.71	0.43	0.01
10	38.26	37.9	37.84	0.36	0.06
11	38.56	38.1	38.13	0.46	0.03
12	38.7	38.3	38.27	0.40	0.03
13	38.99	38.5	38.55	0.49	0.05
14	39.09	38.7	38.64	0.39	0.06

Table 7 shows that the raw MLX90614 readings have a small positive bias compared with the standard thermometer readings. After applying the calibration equation, the absolute error is reduced for all samples, showing that linear calibration improves the temperature measurement accuracy.

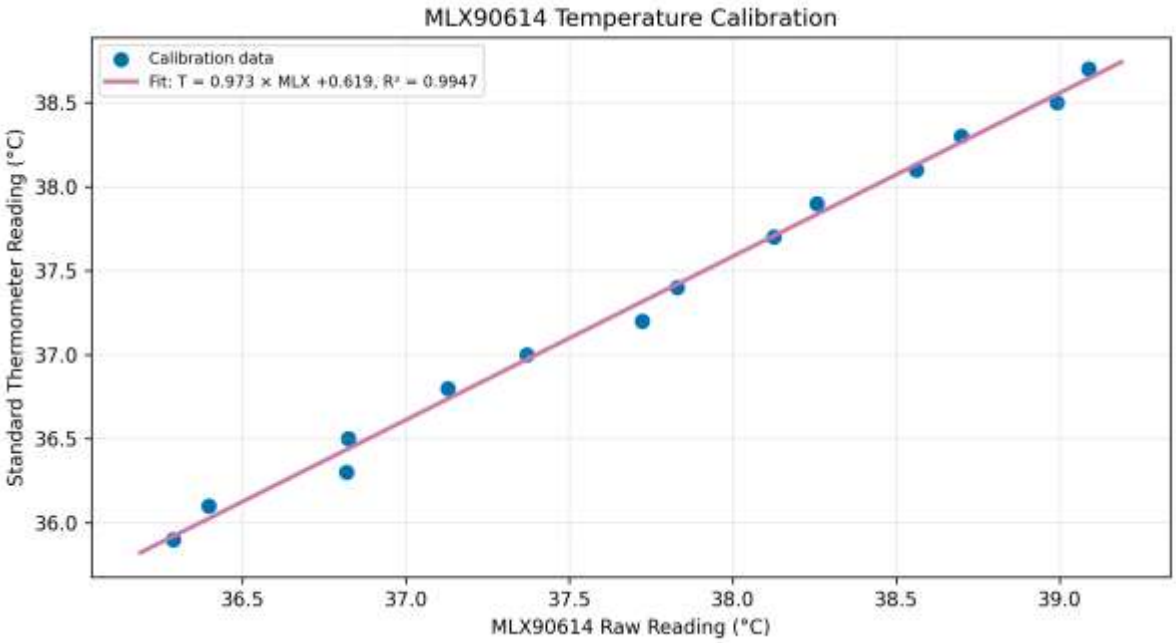


Figure 8: Graph of MLX90614 Vs Standard Thermometer

4.4 IoT Implementation

The ESP32 sent the collected sensor data to the cloud using the MQTT protocol. MQTT was chosen because it is lightweight, fast, and well suited for real-time Internet of Things (IoT) applications. A cloud-based MQTT broker was used to receive and manage the incoming data, which allowed the system to handle real-time sensor streams in a reliable and scalable manner. The stored data was then made accessible to subscribed users for further processing and visualization.

To store the health parameters in real time a web-based dashboard was developed. The dashboard retrieved data from the cloud and continuously updated the displayed values. HTML was used to design the structure of the web page, CSS was applied to improve the visual appearance, and JavaScript was used to fetch data through MQTT and update the interface dynamically.

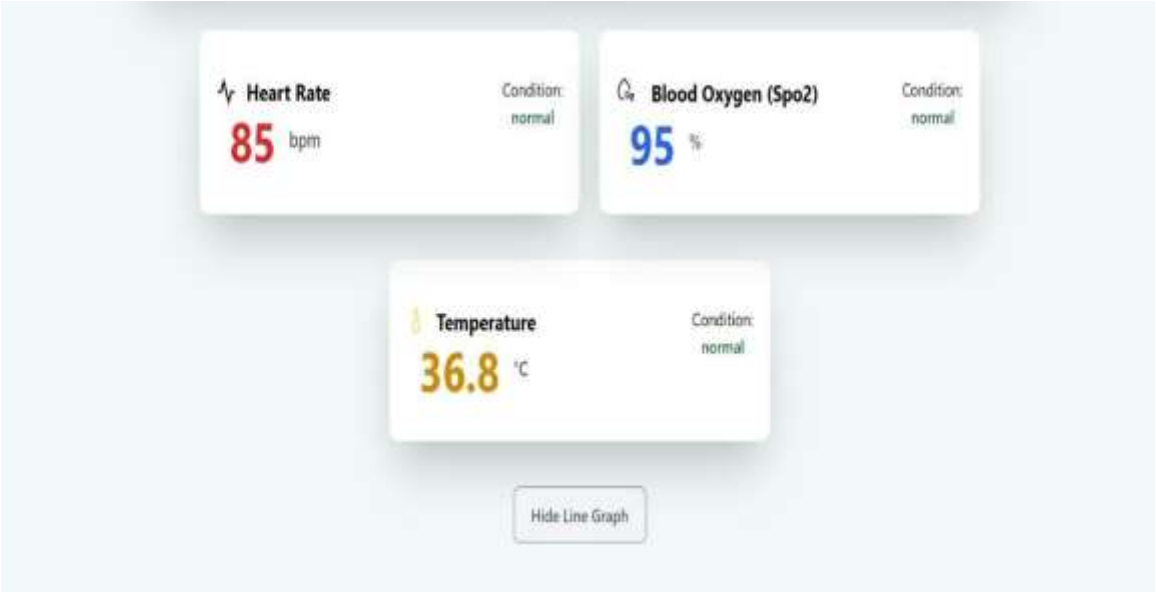


Figure 9: Real Time Data Display

4.5 3D Design

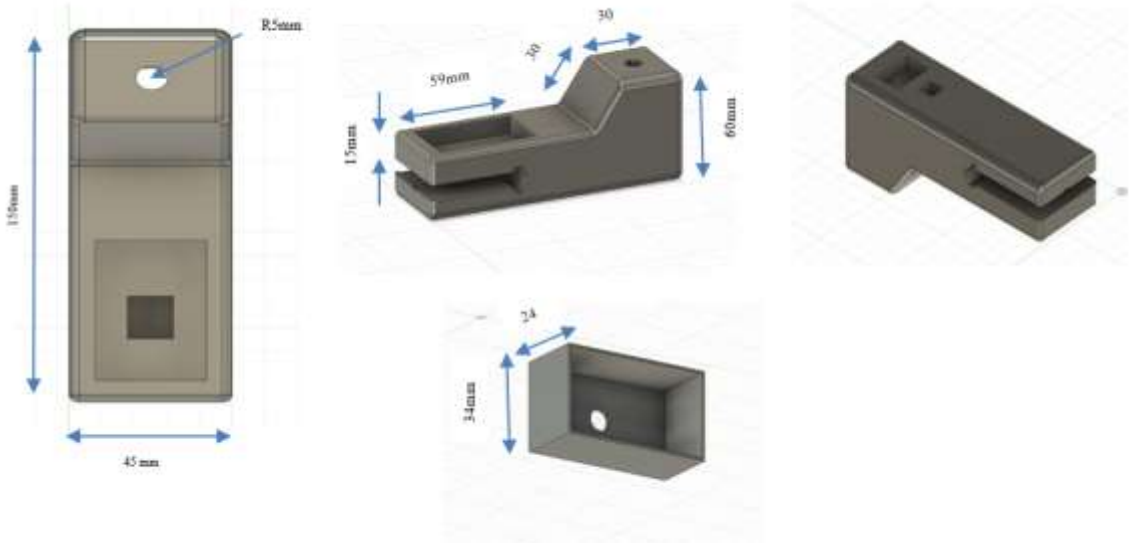


Figure 10: CAD Design

The CAD model represents the mechanical enclosure of the proposed thermo-oximeter device that is designed to integrate both temperature sensing and blood oxygen saturation measurement. The raised back section is used to house the main electronics and temperature sensor with an opening provided for the accurate temperature measurement. The front extended part contains a cavity where the user places their finger for SpO_2 measurement. The top surface includes cut-outs for a display (OLED) and control button.

4.6 Final Design

The final thermo-oximeter device was assembled using a 3D-printed enclosure designed to house all components in a compact and organized manner. The internal section of the enclosure accommodates the main electronics, including the ESP32 module, along with the MLX90614 temperature sensor positioned near a dedicated opening to enable accurate non-contact temperature measurement.

The front section of the device is designed as a finger slot, where the MAX30100 sensor is placed to measure heart rate and blood oxygen saturation (SpO_2) when the user inserts their finger. The top surface of the enclosure includes openings for the OLED display and a push button, allowing real-time visualization of readings and enabling the user to switch between temperature measurement mode and pulse oximeter mode.

For power supply, a 9V battery is integrated within the casing, and a DC-DC buck converter is used to step down the voltage to 3.3V, ensuring proper operation of the sensors and microcontroller. All components are securely fixed within the enclosure, resulting in a portable and user-friendly device capable of real-time data display.



Figure 11: Assembled 3D Thermo-oximeter

4.7 Problems Encountered

4.7.1 Minor Problems

The BPM and SpO_2 readings take a few seconds to stabilize after placing the finger on the sensor. This is because the sensor needs time to gather consistent data.

The readings may fluctuate if the finger is not placed steadily or correctly. Movement or improper contact with the sensor affects light absorption, resulting in inaccurate or unstable values.

Accuracy of thermometer sensor decreases when distance of forehead from sensor's field of view increases. I²C communication delays or data glitches occur, especially when the bus is shared with other devices.

4.7.2 Major Problems Problem

The MAX30100 sensor requires both 1.8V and 3.3V to operate, 1.8V for the internal chip and 3.3V for the red and IR LEDs. These voltages are provided by two onboard regulators. However, a design flaw exists in the module: three pull-up resistors for the I²C lines (SCL, SDA, INT) are connected to the 1.8V line instead of 3.3V. This cause communication issues, especially when used with microcontrollers that expect a higher logic level for proper I²C communication.

Solution:

To solve this issue, the PCB trace connecting the 4.7kΩ pull-up resistors to the 1.8V line was carefully cut to prevent the signal lines from being pulled to an incorrect voltage. Then, a jumper connection was made using a small copper wire to reconnect the pull-up resistors to the 3.3V line. This modification ensured that the I²C communication lines were pulled up to a voltage level that the ESP32 could reliably recognize as HIGH, allowing stable and proper data transmission.

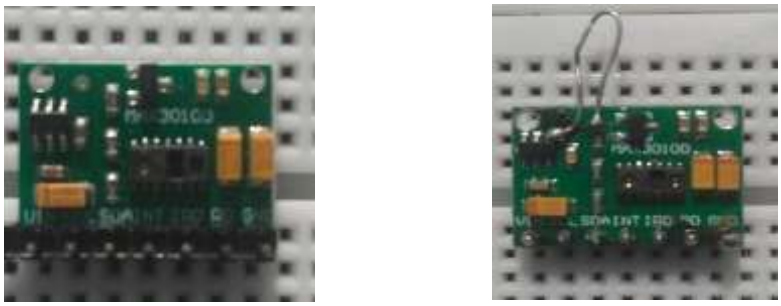


Figure 12: Before and After removing the connection

CHAPTER 5: CONCLUSION

This project successfully designed, fabricated, and implemented a functional Thermo-Oximeter system capable of monitoring heart rate, blood oxygen saturation SpO_2, and body temperature in real time. The system integrates the MAX30100 and MLX90614 sensors with an ESP32 microcontroller, displaying measurements on an OLED screen and transmitting data to a cloud platform via Wi-Fi for remote monitoring.

The final prototype incorporates a 3D-printed enclosure, resulting in a compact, portable, and user-friendly device. The system operates in two selectable modes: pulse oximeter mode and temperature mode, allowing the user to switch between measurements using a push button. The integration of a 9V battery along with a DC-DC buck converter ensures stable power delivery to all components.

The project demonstrates reliable acquisition and display of vital health parameters, along with successful IoT-based data transmission. Overall, the developed device validates the feasibility of combining sensing, processing, and connectivity into a single portable unit.

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